

Zhaodong Liu

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EDUCATION

Carnegie Mellon University – Pittsburgh, PA Master of Computational Data Science (MCDS): Analytics Track	Dec 2027
New York University Shanghai – Shanghai, China Bachelor of Science; Double Major: Computer Science, Mathematics, <i>cum laude</i> Cumulative GPA: 3.793/4.0 CS Major GPA: 3.946/4.0 Math Major GPA: 3.786/4.0 Relevant Coursework: Honors Linear Algebra, Basic Algorithms, Operating Systems, Databases, Natural Language Processing	May 2026

SKILLS

Programming: Python, Java, C, SQL, R | **ML & AI:** PyTorch, LoRA/PEFT, Hugging Face, Transformers
Data & Tools: NumPy, Pandas, MySQL, Git | **Languages:** Chinese (native), English (fluent)

PROFESSIONAL EXPERIENCE

Data Engineer Intern, McDonald's China, Shanghai, China	Dec 2025 - Apr 2026
• Spearheaded an internal AI agent for self-service data access, routing natural-language queries to constraint-aware SQL over a catalog spanning 14 business domains, or RAG over internal data maps and documents to generate cited answers. • Initiated and designed a similarity-based sales forecasting algorithm over a combo master-data set normalized across 4 ID levels, quantifying menu-composition to forecast new-combo launch sales from comparable combos. • Managed and validated enterprise database schemas, governing a 203-scenario and 54-interface data map. • Redesigned data dashboards, organizing 500+ financial metrics with structured drill-down paths for self-service analytics.	

RESEARCH EXPERIENCE

Generative Retrieval for Multimodal Recommendation System - Advisor: Prof. Hongyi Wen Research Assistant, New York University Shanghai, Shanghai, China	May 2025 - Jan 2026
• Developed a multimodal framework for sequential recommendation on Amazon datasets. Identified and applied VQ-VAE, RQ-VAE, OPQ to capture discrete behavior patterns, achieving 47.2% improvement in NDCG@10 (0.0339 to 0.0499). • Collaborated with 2 teammates and implemented BPE-inspired partitioning tokenization across item sequences, merging frequent sub-sequences to enhance contextual coherence and representation quality.	
Energy-Aware Route Optimization Under Traffic-Light Timing - Advisor: Prof. Zhibin Chen Research Assistant, New York University Shanghai, Shanghai, China	May 2024 - Aug 2024
• Proposed and designed a traffic-aware routing system under real-world map constraints, minimizing energy consumption. • Built an OSM, SUMO, TraCI pipeline and extended Dijkstra's algorithm to multi-objective (time, energy) shortest paths, integrating electric and gasoline vehicle energy models. Achieved 10–15% fuel savings at 2–5% time cost.	

PROJECTS

Efficient Retrosynthesis with Pretrained LM Fine-Tuning - Course Project Center for Data Science, New York University, New York, NY	Sep 2024 - Dec 2024
• Trained predictive models for chemical reaction retrosynthesis using SMILES notation, applying an LLM (LlaSMol-Llama2) for dataset reconstruction and augmentation, improving Top-1 Accuracy by 26.2% (0.588 to 0.742). • Explored and employed multiple fine-tuning strategies (encoder freezing, MLP adapters, LoRA) to identify optimal approach with best accuracy-efficiency trade-off.	
Fantasy Sports League Database - Course Project Tandon School of Engineering, New York University, Brooklyn, NY	Sep 2024 - Dec 2024
• Self-directed a project to design a 13-table MySQL database populated with 390+ records, supporting multi-league sports with user authentication, draft scheduling, roster management, and trading systems. Engineered a Python backend and HTML/CSS frontend with ACID-compliant transactions, triggers, and stored procedures, integrating full-stack functionality.	

HONORS

New York University Shanghai, Founder's Day Award	May 2026
New York University Shanghai, Dean's List	Sep 2022 - May 2025
Mathematical Contest in Modeling, Certificate of Achievement, Honorable Mention	May 2023

LEADERSHIP

Trombone Section Leader, Chamber Orchestra, New York University Shanghai	Sep 2022 - May 2026
• Led and trained trombone section, coordinating weekly rehearsals and semester concerts, and oversaw instrument logistics.	
Treasurer, PEER-Pal Club, New York University Shanghai	May 2023 - Aug 2024
• Managed financial operations and budgeting for a club of 100+ club members, mentored incoming club leaders, ensuring continuity of club's charity work; facilitated letter exchanges with 60+ high school students from Hunan Province.	